Pranshu Malik

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Education

Okinawa Institute of Science and Technology Okinawa, Japan
Ph.D. in Computational Neuroscience *Jan. '25 — May '26*

- Withdrew from the program in good standing to better pursue personal research interests and career goals
- Selected coursework: Neuromotor Control, Nonlinear Dynamical Systems, Brain Computation, Cognitive Neurorobotics

University of Western Ontario London, Canada
M.Sc. in Neuroscience; Grade: A *Sept. '22 — Oct. '24*

- Thesis title: The Nature of Reflexes in Online Planning and Control [link]
- Recipient of the Vector Scholarship in Artificial Intelligence (C$17.5k) and the BrainsCAN Graduate Studentship (C$50k)
- Selected coursework: Principles of Neuroscience, Reinforcement Learning

University of Toronto Toronto, Canada
B.A.Sc. in Electrical Engineering; cGPA: 3.92/4.0 *Sept. '17 — Apr. '22*

- Engineering International Scholar: received a full tuition fee waiver for the entire duration of the program (C$229k)
- Recipient of the Adel S. Sedra Gold Medal for achieving the highest cumulative average in the graduating class
- Graduated with High Honors and Minor in Robotics and Mechatronics; Dean's Honor List in all semesters
- Selected coursework (*graduate-level): Linear Control Systems, *Random Processes, *Sensory Communication, Analog and Digital Electronics, System Mapping, Machine Learning, Real-time Control, Robot Modeling and Control, Mechatronics

Work Experience

Graduate Student, OIST Graduate University Okinawa, Japan
Biological Physics Theory Unit and Information Theory, Probability, and Statistics Unit *Jan. '25 — May '26*

- Focusing on understanding closed-loop neural dynamics through behaviorally informative decomposition of control
- Completed several mini projects on topics like adaptive output chunking in recurrent neural networks, passive-active optimal control, dynamical systems reconstruction, speed-curvature power-law, and I/O stability of plastic spiking neural networks

Graduate Student, Western Institute for Neuroscience London, Canada
Sensorimotor Superlab *Sept. '22 — Oct. '24*

- Investigated the coupling between feedback and voluntary motor control by carefully estimating the update times of both modalities; also tried investigating the reliance of implicit adaptation on motor variability
- Led multiple discussions on motor planning and control; also mentored and trained undergraduate research assistants

Navigation Engineering Intern, Zebra Technologies Mississauga, Canada
Path-Planning and Control Team *May '20 — Aug. '21*

- Critiqued, implemented, and tested auto-navigation algorithms from literature for deployment on inventory scanning robots
- Redesigned path planner to improve aisle scan coverage and efficiency by robustly handling obstructions and curved aisles
- Developed tools for rapidly prototyping planners and controllers and benchmarking their performance; written in Julia
- Reviewed and fixed real-world behavior and performance bugs; identified, proposed, and applied planner improvements

Software Engineering Intern, Rocscience Inc. Toronto, Canada
Geotechnical Software Tools Design *May '19 — Aug. '19*

- Developed interactive geometry import and diagnostic tools for file formats external to, and also within, the software suite

Undergraduate Research Assistant, Motion Adaptation Science Lab Toronto, Canada
Toronto Rehabilitation Institute *May '18 — Aug. '18*

- Created finite element models of the lower leg from MRI data for optimizing functional electro-stimulation strategies

Technical Interests

Neuro-inspired Robotics · Systems ID and Integration · Generalizable Control · Perception · Embodied Intelligence

Skills

Python · Julia · Matlab · C/C++ · ROS · PyTorch · OpenCV · Typst · Graphics Design · Mechatronics · Machining

Professional Activities

Conference Poster (CRSy, 2025) · Invited Talk (IIT Gandhinagar, 2024) · Student Leadership & Service (throughout)